KINETIC AND POTENTIAL ENERGY WORKSHEET

Name:_____

Determine whether the objects in the following problems have kinetic or potential energy. Then choose the correct formula to use: $E_k = \frac{1}{2} mv^2$ *or* $E_p = mgh$

1. You serve a volley ball with a mass of 2.1 kg. The ball leaves your hand with a speed of 30 m/s. The ball has _____ energy. Calculate it.

2. A baby carriage is sitting at the top of a hill that is 21 m high. The carriage has mass of 25 kg. The carriage has _____ energy. Calculate it.

3. What work was done to get the carriage up to the top of the hill in question 2?

4. A car is traveling with a velocity of 40 m/s and has a mass of 1120 kg. The car has _____ energy. Calculate it.

5. What work would have to be done by the breaks in order to stop the car in question 4?

6. There is a bell at the top of a tower that is 45 m high. The bell weighs 190 N. The bell has _____energy. Calculate it.

7. You are on roller blades on top of a small hill. Your potential energy is equal to 1,000.0 Joules. The last time you checked your mass was 60.0 kilograms.

   a. What is the height of the hill?

   b. If you start skating down this hill, what will be your speed at the bottom of the hill?

8. Answer the following:

   (a) What is the kinetic energy of a 1 kilogram ball that is thrown into the air with an initial velocity of 30 m/sec?

   (b) How much potential energy does the ball have when it reaches the top of its ascent if it doesn't lose any energy?

   (c) How high into the air did the ball travel?